

STRATEGIC PERFORMANCE & DECISION ANALYTICS DASHBOARD

Project Report

Business Intelligence Analysis — Tableau

Demonstration Dataset: 16 Years of Historical Match Data (2008–2024)

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1. Project Overview

This project is an interactive, two-page Business Intelligence dashboard built in Tableau. Rather than presenting descriptive sports statistics, it treats 16 years of historical match data as an operational dataset — the same way a BI analyst would treat retail transactions or financial records — to demonstrate KPI engineering, relational data modeling, and evidence-based decision support.

The dataset (IPL match and ball-by-ball records, 2008–2024, sourced from Kaggle) was chosen specifically because it is a large, messy, real-world relational dataset well suited to demonstrating BI engineering skills: inconsistent naming across years, a match-centric structure that required restructuring for team-level analysis, and enough scale (260,000+ transaction records) to make data preparation genuinely necessary rather than trivial.

2. Business Problem

Strategic decisions in high-stakes environments are frequently driven by assumption and conventional wisdom rather than empirical evidence. This project set out to test several widely held assumptions against 16 years of historical data:

- Winning the coin toss provides a meaningful competitive advantage.
- Chasing a target is always the safer strategic choice.
- Teams inherently perform better at home venues.
- Higher output (total runs scored) translates directly into a higher win rate.

The objective was to build a centralized, interactive analytical tool capable of testing each assumption objectively, using conditional probability analysis rather than intuition.

3. Dataset

Source: Kaggle — IPL Complete Dataset (2008–2024). Two relational CSV files were used:

- Match-level dataset (dimension table): Match ID, Season, Date, Venue, Team 1, Team 2, Toss Winner, Toss Decision, Match Winner — approximately 1,095 records.
- Ball-by-ball transaction dataset (fact table): Match ID, Batting Team, Bowling Team, Over, Ball, Runs Scored, Wickets — 260,000+ records.

The two tables were joined on Match ID to create a unified analytical dataset spanning 16 IPL seasons.

4. Data Preparation & Cleaning

- Entity resolution: standardized franchise names that changed through rebranding across seasons (e.g., Delhi Daredevils → Delhi Capitals, Kings XI Punjab → Punjab Kings, Royal Challengers Bangalore → Royal Challengers Bengaluru).

- Season format standardization: unified inconsistent historical season formats (2007/08, 2009/10, 2020/21, and plain years) into a single consistent field to support time-series analysis.
- Venue normalization: identified IPL seasons hosted outside India (UAE, South Africa) and classified them as neutral venues, excluding them from home-versus-away advantage calculations to avoid distortion.
- Edge-case handling: matches with no valid result (abandoned or no-result matches) were identified and excluded from win-percentage/efficiency denominators to keep KPIs accurate.
- Unnecessary columns were removed to keep the working dataset focused on analysis-relevant fields.

5. Data Modeling — Team-Centric Restructuring

The source match dataset is match-centric: each row represents one match containing two teams (Team 1 and Team 2). This structure makes reliable team-level filtering difficult — a simple filter on “Team = X” misses every match where that team appears in the other column.

To solve this, each match record was transformed into two team-perspective records — one representing the match from each team’s point of view. For example, a match between RCB and CSK produces one row evaluated from RCB’s perspective and a second row evaluated from CSK’s perspective. This restructuring enabled:

- Accurate, reliable team-level filtering regardless of which column a team originally appeared in.
- Home vs. Away analysis from a single team's consistent viewpoint.
- Bat First vs. Chase analysis at the individual team level.
- Conditional win-probability calculations across any filtered subset of teams or seasons.

6. Measures & Calculated Fields

Measure	Logic
Total Matches	COUNTD(Match ID)
Total Runs	SUM(Total Runs)
Average Runs per Match	SUM(Total Runs) / COUNTD(Match ID)
Season (Cleaned)	SPLIT([Season], '/', 1) — standardizes mixed season formats
Toss Correlation	IF [Toss Winner] = [Winner] THEN 'YES' ELSE 'NO' END
Runner-Up Team	IF [Winner] = [Team1] THEN [Team2] ELSE [Team1] END

7. Dashboard Architecture

Page 1 — Executive Performance Overview

Purpose: a high-level, season-filterable summary of league performance and scoring trends.

- KPI Cards: Total Matches, Total Runs, Average Runs per Match, Champion Team, Runner-Up Team, Leading Run Scorer, Leading Wicket Taker — all dynamic to the selected season filter.
- Season-wise Average Runs per Match (line chart): a 2008–2024 time-series tracking scoring inflation across the tournament's history.

- Total Runs by Team (horizontal bar): aggregate batting output by franchise.
- Team Win % — Completed Matches (horizontal bar): wins divided by matches played, a fairer efficiency comparison than raw totals.
- Point Table: a dynamic table of Matches Played, Wins, and Points for the selected season and league stage.

Page 2 — Match & Strategy: Impact on Winning

Purpose: quantify how strategic decisions and venue context influence match outcomes, using conditional win-probability analysis.

- Impact of Toss on Match Result: Win % when the toss winner won vs. lost, filterable by team and season.
- Batting Strategy Impact on Win %: conditional win probability for Bat First vs. Chase.
- Home vs. Away Win %: conditional win probability by venue familiarity, with neutral-venue seasons excluded.
- Team and Season filters apply globally across all three charts on this page, recalculating each conditional probability live.

8. Key Insights & Interpretations

- Output does not equal efficiency: comparing the Total Runs and Win % charts side by side (Page 1) shows that some franchises generate high run volumes without a proportionally higher win rate — production volume and win efficiency are distinct metrics that don't automatically move together.
- Scoring has inflated significantly over 16 seasons: the season-wise trend line shows a clear upward shift in average runs per match, particularly pronounced in the most recent seasons — reflecting an evolving, increasingly aggressive scoring environment.
- Execution outweighs the toss: across the full dataset, teams that won the toss won 50.6% of matches versus 49.4% for teams that lost it — functionally a coin flip. Whatever marginal advantage exists is negligible compared to actual execution.
- Home advantage is real and measurable: 55.7% (home) vs. 47.0% (away) across the full dataset, neutral venues excluded — a consistent 8.7 percentage-point advantage, verified directly by subtracting the two dashboard values.
- Batting strategy matters, but less than assumed: Bat First vs. Chase shows 45.7% vs. 53.9% across the full dataset — chasing holds a real edge, but the gap is more moderate than conventional cricket wisdom often suggests, and narrows or widens depending on the specific team/season filter applied.

9. Technologies & Methods

- Business Intelligence: Tableau Desktop, Tableau Public, interactive dashboard design
- Data Engineering: data cleaning, entity resolution, relational data modeling, team-centric data restructuring
- Analytics: KPI engineering, trend analysis, comparative analysis, conditional probability analysis

10. Conclusion

This project demonstrates how Business Intelligence methodologies — data cleaning, relational data modeling, KPI engineering, and conditional probability analysis — can transform a large, messy, real-world transactional dataset into an interactive tool for testing assumptions against evidence. The specific findings (the toss functioning as a near-exact coin flip, a measurable and quantifiable home advantage, and the divergence between output and efficiency) illustrate a pattern applicable well beyond this dataset: raw volume metrics and conventional assumptions frequently diverge from what conditional, evidence-based analysis actually shows.